

LoRa vs. WiFi Ad Hoc: A Performance Analysis and Comparison

George Klimiashvili, Cristiano Tapparello, Wendi Heinzelman

Department of Electrical and Computer Engineering, University of Rochester, Rochester, NY, USA

Email: gklimias@u.rochester.edu, {cristiano.tapparello, wendi.heinzelman}@rochester.edu

Abstract—With increasing demand for wireless networks, the number of technologies that allow us to connect remote devices has increased as well. One of the more common technologies that allows us to connect two remote devices with each other is WiFi, which provides high power, short-range connection between devices. In order to transfer data using WiFi over a longer distance, it is common to use Mobile Ad Hoc Networks (MANETs), which provides an infrastructureless network that can be used for long range communication in the cases where infrastructure-based networks are unavailable. Another, more recent and less common way to connect remote devices without relying on pre-existing infrastructure is by using LoRa, which provides low power, long-range wireless connection. In this paper, we analyze the performance of WiFi Ad Hoc and LoRa networks using Network Simulator 3 (ns-3) in terms of delay and energy consumption required to send a file of a given size as we vary the distance from the source to the destination. In addition, we develop mathematical models for the delay and energy consumption, and compare these models with the results obtained through simulation. Our results show which network protocol, WiFi Ad Hoc or LoRa, provides the lowest delay or the least energy consumption for a given file size and distance, which can be used to adaptively select the communication protocol to optimize the network performance.

I. INTRODUCTION

WiFi and cellular are two of the most common types of wireless communication networks. Both of these networks rely on pre-existing infrastructure in order to operate. While much of the world has access to some type of infrastructure-based network, especially in urban areas, there are still many areas where deploying an infrastructure-based network is difficult if not impossible. A recent report by OpenSignal examined mobile data performance and availability in 95 countries [1]. The report, which involved 822,556 devices, showed that 3G or better data signal availability varied from 36.50% coverage throughout Guyana to 98.54% coverage throughout South Korea, with 15 out of 95 countries having less than 70% network availability [1]. Moreover, in the cases of natural disasters or wartime, the infrastructure that makes communication possible might be unavailable, which makes the existence of a network that does not rely on pre-existing infrastructure more vital. All of this creates demand for communication protocols that allow devices to directly communicate with each other.

Given the demand for infrastructureless networks, we need to identify which network will be more efficient and more practical for a given application. Depending on the scenario, we might need to transfer a small amount of data over large distances using very little power, or we might need to transfer a

large amount of data as fast as possible, with energy efficiency not being a priority. Given the various applications' settings and requirements, it is essential to analyze which type of communication technology and network protocol represents the best solution for a particular scenario.

In this paper, we compare and analyze the performance of two of the most popular ad hoc network protocols, namely WiFi and LoRa, for different types of application requirements. WiFi Ad Hoc and LoRa are two radically different technologies that allow ad hoc wireless communication between devices. We aim to find which protocol provides a faster and/or a more energy efficient way to transmit data under different scenarios. Specifically, we want to be able to determine the delay and energy consumption required to send a file of a given size to a destination located at a particular distance from the sender when using WiFi Ad Hoc or LoRa. Using Network Simulator 3 (ns-3) [2], we simulate the performance of WiFi Ad Hoc and LoRa for different packet sizes and communication distances, and derive an analytical model of the delay and energy consumption of both protocols to estimate their performance under different settings. As wireless communications are limited in range by the particular radio technology used, multi-hop communications is also considered in our analysis. Our results can be used to determine which network protocol is more effective for a given scenario.

The rest of the paper is organized as follow. In Section II, we provide background on WiFi Ad Hoc and LoRa and discuss some of the related work. In Section III, we present mathematical models of the delay and energy consumption for both protocols. In Section IV, we introduce the simulation environment and data collection mechanism. In Section V, we provide the simulation results and compare them with the mathematical models. Section VI concludes the paper.

II. BACKGROUND AND RELATED WORK

A. Background and Related Work on LoRa

LoRa is a recent Industrial, Scientific and Medical (ISM) based wireless technology that operates in the sub-GHz frequency band and has been proposed by the LoRa Alliance [3]. LoRa uses a Chirp Spread Spectrum (CSS) modulation, in which chirps, sinusoidal signals of linearly varying frequency and fixed duration, are used to encode information [4], [5]. LoRaWAN defines the communication protocol and system architecture of the physical layer LoRa and is designed to optimize Low-Power, Wide-Area Networks (LPWAN) [6].

Table I
DATA RATE FOR LoRa IN EU REGION

Data Rate	Configuration	bit/s	Max Payload [B]
0	SF12/125kHz	250	59
1	SF11/125kHz	440	59
2	SF10/125kHz	980	59
3	SF9/125kHz	1 760	123
4	SF8/125kHz	3 125	230
5	SF7/125kHz	5 470	230
6	SF7/250kHz	11 000	230

According to the specification [6], [7], a LoRa gateway or base station has a communication range of 2-5km in urban areas and up to 15km in suburban areas depending on the spreading factor (SF). The SF, for LoRa, is defined as the binary logarithm of chirps per symbol [4]. The SF determines the data rate used for the communication, which varies from 0.25kb/s to 50kb/s, and the maximum payload size of the packet that can be sent [8], which varies from 59 Bytes to 230 Bytes (see Table I). It is important to note that due to local regulations, LoRaWAN has a duty cycle of approximately 1% for most regions [9], which underlines its applicability for sending small files.

The LoRaWAN network architecture consists of End Devices (EDs), Gateways (GWs) and a Network Server (NS). EDs send and receive messages from one or more GWs, which are used to relay the messages to the NS [5]. For the purpose of our comparison, we will focus our analysis on the communication between EDs and GWs.

The performance of LoRaWAN in an indoor environment was analyzed in [10]. The experimental setup consisted of one ED, one GW, and one NS. In order to analyze the performance of LoRa, the authors conducted a total of 80 hours of measurement, during which packets of 4 B (plus 13 B of headers and MIC) were sent with a delay between packets depending on the data rate that was used in the experiment. The paper found that none of the measured currents varied with a change in the data rate. It was also found that with data rate 6, the maximum data that can be transmitted per day is 1MB, while the same number for data rate 0 is 15 kB. Finally, the paper found that the delay associated with sending the 4 B packet and receiving an acknowledgment message was approximately 2.08s for data rate 6.

The authors in [11], presented an ns-3 module and compared it to a real-world testbed, as well as attempted to improve the performance of a large scale LoRa network consisting of 1 ED and up to 10000 GWs, while preserving the energy consumption per node. The paper found that the ns-3 module approximates the behavior of a real-life LoRa network in an accurate way. Moreover, the authors found that the suggested enhancement - Carrier Sense Multiple Access (CSMA) leads to lower collision ratios while only slightly increasing the energy consumption per node.

B. Background and Related Work on WiFi Ad Hoc

In contrast to LoRa, a WiFi ad hoc network consists of multiple devices that communicate with each other through WiFi.

WiFi allows high throughput, short range communications between devices, and while it is mostly used to implement single-hop networks, it can also be used to implement multi-hop ad hoc networks.

In a multi-hop WiFi ad hoc network, the devices that are within range of each other, which generally varies between 100m and 250m depending on the particular 802.11 WiFi standard used, connect to each other, forming a network that can operate across significantly longer distances. In WiFi ad hoc, the devices use nearby devices as relays in order to share data with a device that is out of their transmission range. The route from the source to the destination is found using a routing protocol. Once the path is found and available to transmit and receive the data, the source device starts sending the file to the next device in the route, which in turn forwards it to the next device in the route until the destination is reached.

The energy consumption of IEEE 802.11 wireless networks operating in ad hoc mode was analyzed in [12]. Based on a series of experiments, the authors presented an analytical model to calculate the energy consumption of a one-hop ad hoc network, using UDP traffic. The model used to represent the energy consumption of sending, receiving or discarding a packet of size s bytes was defined as $E(s) = m * s + b$, where m is the energy required for sending one byte, and b is a fixed cost. The paper found that fixed costs are very high and energy consumption associated with receiving packets cannot be neglected.

Similarly to [12], [13] used a linear model to represent the energy consumption of IEEE 802.11 wireless Ad Hoc networks. Based on simulations in ns-2, the authors compared the performance of two common routing protocols: Dynamic Source Routing (DSR) and Ad Hoc On-Demand Distance Vector (AODV), in order to understand the trade-offs of protocol specific behaviors. The paper found that even though DSR was more effective in terms of utilizing bandwidth, which translated into a higher number of packets sent and received in a given time, it was less efficient in terms of energy consumption when compared to AODV.

The impact of multi-hop communications on a MANET was presented in [14]. In particular, the authors analyzed how the number of hops affects the TCP throughput in a linear, fixed, multi-hop 802.11 network. The results indicated that the throughput dropped from 1463 Kbps in the case of a 1 hop network to 246.4 Kbps in the case of a 5 hop network, to 182.4 Kbps in the case of a 10 hop network.

III. SYSTEM MODEL

The goal of our work is to compare the delay and energy consumption of LoRa and WiFi Ad Hoc associated with sending a file of different sizes over different distances. To this end, we consider two different network topologies: a linear network composed of N nodes, with a distance d between adjacent nodes and a random network consisting of M nodes in which the source and the sink are separated by a fixed distance m , while the remaining nodes are randomly distributed in an $m \times n$ area. As shown in Figure 1, the leftmost

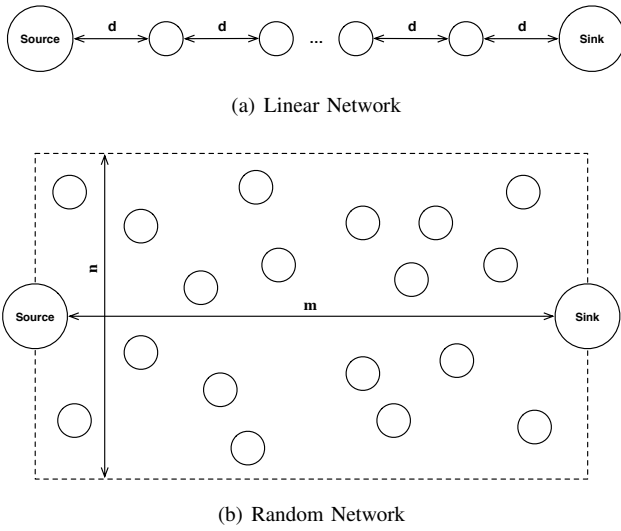


Figure 1. Considered network topologies.

node acts as a source and sends a message to the sink node. To better replicate real scenarios, we also consider additional flows between randomly selected nodes, to evaluate the impact of interference to the primary flow.

On the other hand, thanks to the long range capabilities offered by the LoRa protocol, in the rest of the paper we assume that when using LoRa, the source can directly communicate with the sink. Due to the 1% duty cycle of LoRa, small offsets between concurrent communications reduce the mutual interference [15]. Therefore, for the purpose of our comparison, we will not be taking it into consideration.

In what follows, we model the delay and energy consumption required to send a file from the source node to the sink, using either LoRa or WiFi Ad Hoc with a linear topology. While the linear topology may be an unrealistic assumption for many real life networks, it allows us to find lower bounds on the delay and energy consumption for a generic WiFi Ad Hoc network. While these models do not aim to represent the detailed operation of WiFi Ad Hoc and LoRa, they provide a relatively accurate representation of the corresponding delay and energy consumption and trade-offs associated with each type of network.

A. Delay Model

In order to calculate the delay associated with sharing a file of a given size using WiFi Ad Hoc, we use a linear model. According to our model, the delay associated with sharing a file of size s over a linear network of n hops with a payload size of S is given by:

$$D_{WF}(n, s, S) = n \left(\frac{S + H}{T_{WF}} \left\lceil \frac{s}{S} \right\rceil + \beta \right) + \alpha \quad (1)$$

where T_{WF} is the link throughput, β is a fixed delay associated with sharing the file, α is a fixed delay required to initiate the data sharing, $\lceil s/S \rceil$ is the number of packets sent and $H = 64$ B is the total size of the IP, UDP and MAC headers. We

note that Eq. (1) assumes that the throughput is constant and does not depend on the file size or any other characteristic of the network, and does not take into consideration routing and queuing delay, interference, or advanced pipelining.

Contrary to the model for WiFi Ad Hoc networks, the delay model of a LoRa communication cannot be represented in a simple manner, due to the duty cycle and maximum payload size limitations of the standard. In particular, sending a file of large size creates the need to packetize the file into smaller packets. The delay of sending each of these packets depends on the size of these packets, as LoRa provides relatively constant throughput, independent of the packet size or any other characteristics. Thus, we first model the delay associated with the transmission of a file of size s B, when using a data rate with maximum payload size S as:

$$D_{LR}^{tx}(s, S) = \left\lfloor \frac{s}{S} \right\rfloor \left(\frac{S + H}{T_{LR}} + \gamma \right) + \frac{s \bmod S + H}{T_{LR}} + \gamma, \quad (2)$$

where $\lfloor s/S \rfloor$ is the number of packets of size equal to the maximum payload, $H = 9$ B is the size of the header attached to each packet, T_{LR} is the link throughput, $s \bmod S$ is the residual number of bytes that do not fit in a full packet, and γ is a fixed time constant.

Due to the duty cycle defined by the standard, when the transmission of the file requires multiple transmissions, the ED needs to switch to Idle state for a duration proportional to the duty cycle D . In particular, when sending M packets of size less than or equal to S , the total idle time is given by

$$D_{LR}^{id}(M, S) = (M - 1) D_{LR}^{tx}(S, S) \frac{(1 - D)}{D}. \quad (3)$$

where $M - 1$ is the number of times the device will enter the idle state, and $D_{LR}^{tx}(S, S) \frac{(1 - D)}{D}$ is the length of each idle state. Therefore, by combining Eqs. (2) and (3), the total delay required for sending a file of generic size s is given by

$$D_{LR}(s, S) = D_{LR}^{tx}(s, S) + D_{LR}^{id} \left(\left\lceil \frac{s}{S} \right\rceil, S \right). \quad (4)$$

Given the above, Eq. (4) implies that the delay required to send a large file is given by the sum of the time that is required to send as many packets as possible with maximum payload size, the time that the devices will have to spend in idle state after sending each of these packets, and the time that is required to send the last packet that has a size less than or equal to the maximum payload size.

B. Energy Model

During the process of sharing a file, each node in the WiFi Ad Hoc network switches between three states: Idle, Transmission and Reception. Because of the nature of the ad hoc network, at any point in time, transmission occurs between only one pair of devices, which means that one device sends the data, one receives the data and all the rest of the nodes are in idle state. Assuming there is no interruption in the transmission, we can write the energy consumed by the WiFi Ad Hoc network as:

$$E_{WF}(n, s, S) = D_{WF}(n, s, S) [P_{WF}^{tx} + P_{WF}^{rx} + (n - 1) P_{WF}^{id}] \quad (5)$$

Table II
DEFAULT PARAMETERS OF THE LoRa ENERGY MODEL [16]

Idle (μA)	Sleep (μA)	TX (mA)	RX (mA)
1.5	0.2	120	11.5

where P_{WF}^{tx} is the power consumption of the node during transmission, P_{WF}^{rx} is the power consumption required to receive a packet, P_{WF}^{id} is the idle power, and $n - 1$ is the number of devices that are in IDLE state. Once again, this formula does not take into consideration the cost associated with routing or queuing or any potential interference.

Similarly, in order to calculate the energy consumption of LoRa, we need to consider LoRa's duty cycle. In particular, the energy consumption of LoRa can be calculated as follow:

$$E_{LR}(s, S) = D_{LR}^{tx}(s, S)(P_{LR}^{tx} + P_{LR}^{rx}) + 2D_{LR}^{id}\left(\left\lceil \frac{s}{S} \right\rceil, S\right)P_{LR}^{id}, \quad (6)$$

where P_{LR}^{tx} is the power consumption of the LoRa node during transmission, P_{LR}^{rx} is the power consumption required to receive a packet, and P_{LR}^{id} is the idle power.

The above equation implies that the total energy consumed is equal to the sum of energy consumed by the ED for sending the data, the energy consumed by the GW for receiving the data, and the energy consumed by both when they are in idle state due to the duty cycle.

IV. SIMULATION ENVIRONMENT AND CONTRIBUTIONS

In order to analyze the delay and energy consumption of a wireless network using LoRa and WiFi Ad Hoc, we simulate their performances using ns-3 [2], as it provides a reliable and easy to use platform to simulate wireless networks.

While simulation of the LoRa network is not natively supported by ns-3, an ns-3 module that implements the LoRa specifications has recently been released by Davide Magrin [5]. This module includes all the functionalities required to simulate LoRaWAN networks, such as classes that mimic the behavior of EDs and GWs. Moreover, the module implements aspects of all the different network layers, from the physical (PHY) to the application layer. In addition, the model includes the duty cycle limitations described in the LoRa standard, as well as the signal loss caused by different types of interference [5].

While the LoRaWAN ns-3 module implemented by Davide Magrin includes most of the tools necessary to simulate the LoRa network, it does not include the complete energy model that would allow us to calculate the energy consumption of both the ED and GW. To this end, we extended the module to include an energy model for LoRa networks, and an application sender that allows us to reach maximum throughput given LoRa's duty cycle and maximum payload size limitations.

Our *LoRa Energy Model* calculates the total energy consumed during the simulation by each of the nodes. The default current values used in the energy model are given in Table II. The total energy consumed by each node is calculated following the ns-3 energy model specifications: after a change

Table III
WiFi AD HOC PARAMETERS

Protocol	802.11g
Data Rate (Mbps)	54
Transport Protocol	UDP
Payload Size (B)	512
Propagation Loss Model	Log-distance Propagation Loss Model
Path Loss Exponent	3
Reference Loss at 1m (dB)	40.04
Routing Protocol	OLSR
Hello Interval (s)	0.5
TC Interval (s)	1
TX Gain (dB)	0
RX Gain (dB)	10

of state occurs, the energy consumed while being in that state is calculated by multiplying the time spent in the state by the power consumption of the node in that state, and this energy is then added to the total energy consumed so far by the node. Thus, the total energy consumed is calculated by adding up the total energy consumed while the device is in each communication state.

In order to be able to send a file that is larger than the maximum payload of LoRa, we designed an application sender that takes into consideration the duty cycle limitations of LoRa. The application sender divides files into packets of 230 B, which is the maximum payload size for the highest data rate as defined by the standard (see Table I), and sends the packets in a periodic manner, taking into consideration the sleep time associated with the 1% duty cycle of LoRa. This application sender allows us to reach the maximum throughput at which the LoRa network can operate.

Our extension to the LoRaWAN ns-3 module can be downloaded from [17].

V. NUMERICAL RESULTS

In this section, we present ns-3 based simulation results for LoRa and WiFi Ad Hoc, as well as a comparison of the simulation results with the mathematical models described in Section III. We show the delay and energy consumption associated with using each type of network in order to send a file of a given size over a given distance.

A. Simulation Parameters

The parameters for WiFi Ad Hoc are given in Table III. Note that the path loss exponent of 3 represents a relatively lossy environment such as urban area, and reference loss value of 40.04 dB is based on the free-space path loss (FSPL) formula. OLSR was selected as the routing protocol, which is natively implemented in ns-3. "Hello Interval" and "TC Interval" given in Table III determine the periodicity of Hello and Topology Control (TC) Messages. Further information about the purpose and impact of the Hello and TC messages can be found in [18]. We note that the choice of "Hello Interval" and "TC Interval" is crucial for the purpose of our experiment, as smaller values lead to a faster finding of the route, however, they also lead to the data transmission being interrupted more often,

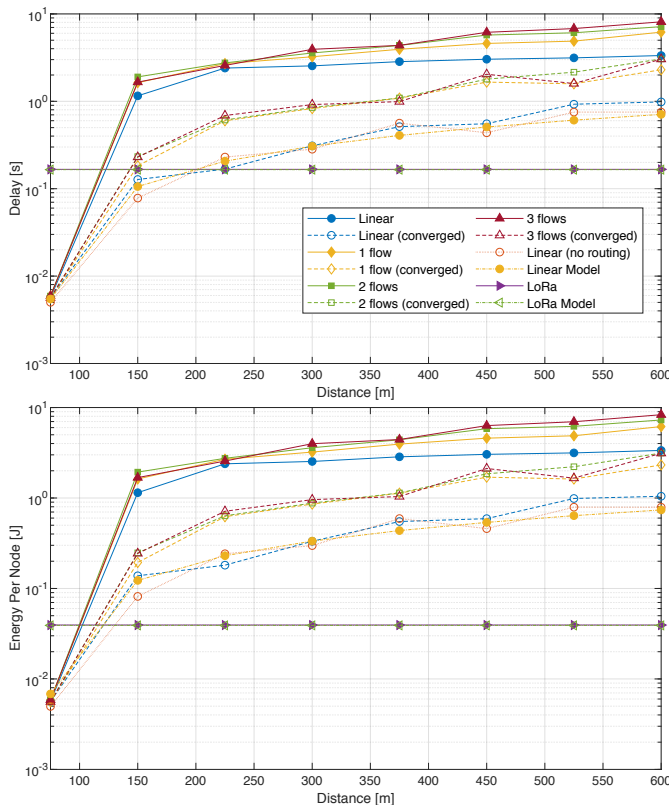


Figure 2. LoRa vs WiFi Ad Hoc: Delay and Energy consumption per node for variable distance, file size = 200 B.

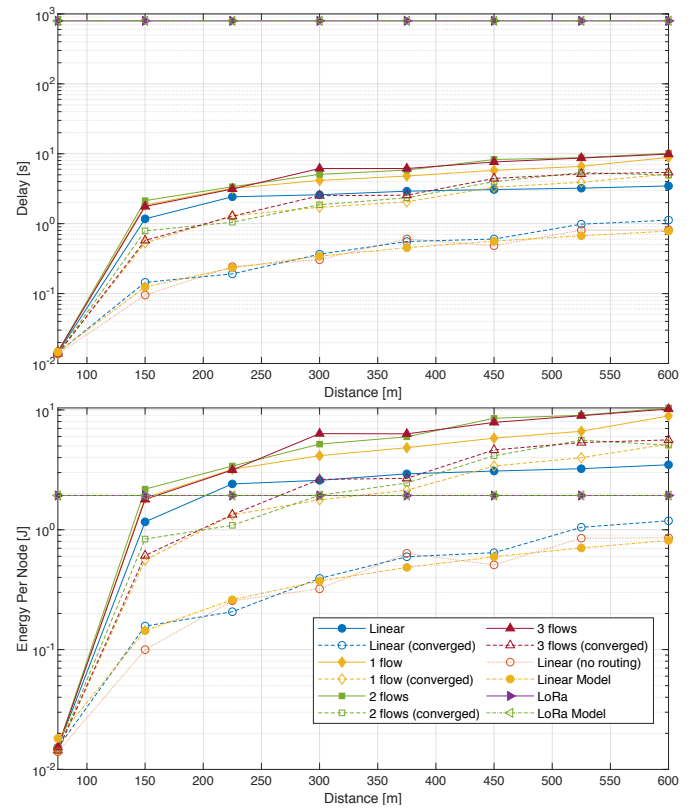


Figure 3. LoRa vs WiFi Ad Hoc: Delay and Energy consumption per node for variable distance, file size = 10 kB.

increasing the delay. Due to the fact that we are dealing with large networks and small file sizes, we chose the previously mentioned values, to ensure fast routing.

For the simulations of the LoRa protocol, we considered a network consisting of one ED (source), one GW (sink), and one NS. The measurements of delay were taken for sending a file between the ED and the GW, and the total energy consumed was calculated by adding up the energy consumed by the ED and the GW. We used the Log-distance Propagation Loss Model with pass loss exponent of 3, reference distance of 1 m and reference loss value of 31.23 dB, based on FSPL formula. Furthermore, we used data rate 6, which theoretically can reach a throughput of 11 Kbps, has a SF of 7 and has maximum payload size of 230 B (see Table I).

In order to compare the performance of LoRa and WiFi Ad Hoc networks, we consider four different scenarios:

- 1) Variable distance with small file size;
- 2) Variable distance with large file size;
- 3) Variable file size over a small distance; and
- 4) Variable file size over a large distance.

For each of these scenarios, in the case of WiFi Ad Hoc, we use linear and random topologies. In the linear topology, the distance between adjacent nodes is 75 m. In the random topology, the nodes are randomly distributed in an l by $l/10$ rectangle with approximately 1 node per 450 m², where l is the distance between the source and the sink. In the case of LoRa,

the ED directly communicates with the GW, thus eliminating the need for exploring different topologies.

Moreover, in order to demonstrate the effect of interference on the network, we simulate scenarios with multiple flows between randomly selected nodes, where in Figures 2-5 “1 flow” represents no additional flows, “2 flows” represents one flow in addition to the source-sink flow, and “3 flows” represents two flows in addition to the source-sink flow. As mentioned before, due to the 1% duty cycle of LoRa, interference between concurrent flows can be easily avoided [15].

Finally, for WiFi Ad Hoc, we consider both converged and non converged networks depending on whether the routing table at each node contains information about the entire network (denoted as “converged” in Figures 2-5). To completely eliminate the overhead introduced by the routing protocol, we also consider a linear network in which we manually set the routing tables. These scenarios give further insight into trade-offs associated with routing in WiFi Ad Hoc and how these trade-offs affect its performance compared to LoRa.

Based on these scenarios, we aim to find which protocol will be more efficient in terms of delay and energy consumption. All the results presented in this section are the average of 50 independent runs over channel realization and nodes’ position.

B. Impact of Distance

In order to evaluate the impact of distance on the performance of the protocols, we vary the distance between the

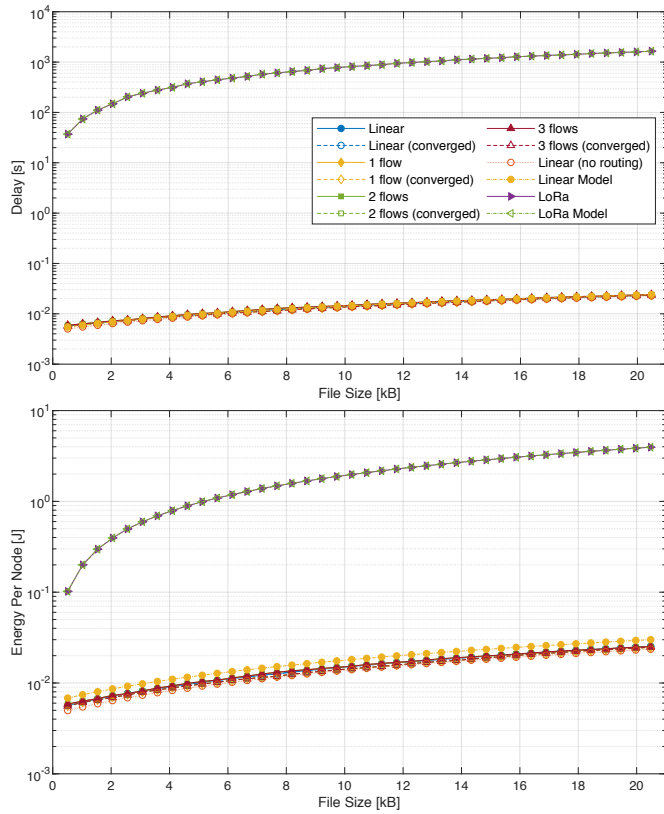


Figure 4. LoRa vs WiFi Ad Hoc: Delay and Energy consumption per node for variable file size, distance = 75m.

source and the sink from 75 m to 600 m, with a step of 75 m. According to the system model described in Section III, for LoRa this means moving the ED and GW away from each other, while for WiFi Ad Hoc we increase the number of hops for the linear topology and we increase the area in which nodes are distributed for the random topology. For example, for covering a distance of 300 m, the linear WiFi Ad Hoc network requires 5 nodes placed $d = 75$ m apart, while random topology requires 21 nodes distributed in 300 by 30 rectangle.

Figure 2 shows the simulated and modeled delay and energy consumption per node required to send a file of size 200 B, which is close to but still less than the maximum payload size of LoRa for the highest data rate. The results indicate that for such file sizes, LoRa has lower delay, for distances greater than 225 m or networks exceeding 3-hops and lower energy consumption, for distances greater than 75 m or distances that require WiFi Ad Hoc to use multiple hops. This is because, even though LoRa has lower throughput, it does not require any time for routing or forwarding the messages, which decreases the delay significantly when compared to a WiFi Ad Hoc network. Along with the low power nature of LoRa, this low delay leads to significantly lower energy consumption. Based on these results, we can see that both LoRa and WiFi Ad Hoc delay and energy models approximate well the simulation results when the overhead of routing is not considered.

Figure 3 compares the performance of LoRa and WiFi Ad

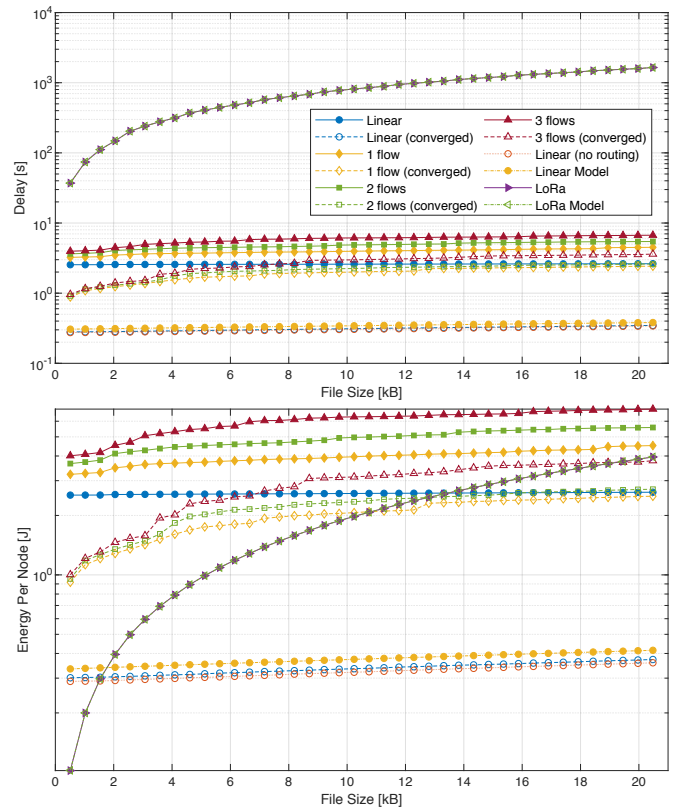


Figure 5. LoRa vs WiFi Ad Hoc: Delay and Energy consumption per node for variable file size, distance = 300m.

Hoc when sending a file of size 10 kB, which is greater than the maximum payload size defined in the LoRa standard, therefore requiring the LoRa ED to enter the sleep mode multiple times. The results indicate that for large file sizes, the delay associated with sending these large files is significantly higher for LoRa than it is for WiFi Ad Hoc, for any topology and interference. Such a big gap in the delay is due to two factors: LoRa's 1% duty cycle and very low throughput. In addition, the results from Figure 3 show that while WiFi Ad Hoc outperforms LoRa in terms of energy consumption per node for short distances (less than 300 m for random converged networks), LoRa is more efficient for larger distances. This is because for small distances, WiFi Ad Hoc uses few hops and few devices, which leads to smaller forwarding overhead and faster sharing of data, which in turn leads to lower energy consumption. For large distances, instead, the number of nodes and hops in WiFi Ad Hoc increases, which implies that data needs to be forwarded more times therefore increasing both the delay and the energy consumed by each node. On the other hand, the delay and energy consumed by LoRa do not depend on the distance between source and sink, as LoRa devices are able to communicate over large distances (up to 15 km [6]). Because of the low power nature of LoRa, the energy consumed by the network is relatively small. Finally, we can see that LoRa and WiFi Ad Hoc delay and energy models approximate the ns-3 simulations very accurately.

C. Impact of File Size

In Figures 4 and 5, we fixed the distance between the source and the sink to 75 m and 300 m, respectively, and varied the file size from 512 B to 20.48 kB with 512 B steps. We note that a distance of 75 m implies that the WiFi Ad Hoc network only requires a single hop, while the distance of 300 m implies that the WiFi Ad Hoc network requires 5 nodes or 4 hops in the linear topology and 21 nodes in the random topology. This implies that there will be a significant increase in both delay and energy consumption due to the cost associated with routing the packets, when compared to 75 m.

The delay and energy consumption per node of LoRa and WiFi Ad Hoc with variable file size over a fixed distance of 75 m are illustrated in Figure 4. The results indicate that for short distances over which WiFi Ad Hoc requires only a single hop, WiFi Ad Hoc outperforms LoRa in both energy consumption and delay for any file size. These results are due to the fact that when WiFi Ad Hoc is operating using a single-hop, it does not require any time for routing, which significantly decreases both delay and energy consumption. Moreover, because the theoretical throughput of WiFi is almost 5000 times higher than that of LoRa's, as well as because of the LoRa duty cycle, the time required to send a packet becomes significantly smaller for WiFi Ad Hoc as the file size increases. Consequently, significantly lower delay by itself leads to lower energy consumption for WiFi Ad Hoc. Finally, while the energy and delay model of LoRa and the delay model of WiFi Ad Hoc fit the simulation results relatively well, we can see that the energy model of WiFi Ad Hoc slightly overestimates the simulation results. One of the possible reasons for this is the assumption that we made that there are no interruptions between the transmission and reception of multiple packets. In ns-3 simulations, nodes switch from TX and RX states after they send or receive each packet, which means that our assumption that at any point in time there is one node that is in the TX state and there is one node that is in the RX state is not completely accurate. However, as we can see throughout different scenarios, our simplified energy model provides relatively accurate prediction about the energy consumed in WiFi Ad Hoc network.

Figure 5 shows the energy and delay required to send a file of variable size, over a fixed distance of 300 m between the source and the sink. The results indicate that while WiFi Ad Hoc is always more efficient in terms of delay, LoRa outperforms converged WiFi Ad Hoc with random topology in terms of energy consumption for file sizes up to 11 kB. The significantly higher delay provided by LoRa is once again caused by its duty cycle limitation. In terms of energy consumption, LoRa performed better than WiFi Ad Hoc because of the low power nature of LoRa, as well as due to the fact that a large number of communication hops in WiFi Ad Hoc leads to longer delay, which implies that more energy is consumed by each node. Finally, we can see that energy and delay models for LoRa and WiFi Ad Hoc approximate the simulation results relatively accurately.

VI. CONCLUSIONS

In this paper, we analyzed and compared the performance of a WiFi Ad Hoc network and LoRa in terms of energy consumption and delay required to send a file from a source node to a remote sink. According to our simulation results, and our proposed mathematical models, we are able to conclude that WiFi Ad Hoc performs better in terms of delay for files that are larger than the maximum payload size of LoRa. Moreover, WiFi Ad Hoc is more energy efficient when used over small distances that can be covered using a single hop. On the other hand, LoRa tends to be more energy efficient when transferring files over distances that exceed the maximum range of WiFi. Finally, LoRa provides lower delay when sending small packets over distances that exceed the operation range of WiFi. Future work includes a comparison of the protocols on real devices, and the development of protocols that dynamically change the wireless technology in use to optimize the network performance.

REFERENCES

- [1] OpenSignal, "Global state of mobile networks," <https://opensignal.com/reports/2016/08/global-state-of-the-mobile-network>, accessed: July 2019.
- [2] "Network simulator 3 (ns-3)," <https://www.nsnam.org>, accessed: July 2019.
- [3] A. J. Wixted, P. Kinnaird, H. Larijani, A. Tait, A. Ahmadinia, and N. Strachan, "Evaluation of LoRa and LoRaWAN for wireless sensor networks," in *IEEE SENSORS*, 2016, pp. 1–3.
- [4] A. Augustin, J. Yi, T. Clausen, and W. Townsley, "A study of lora: Long range & low power networks for the internet of things," *Sensors*, vol. 16, no. 9, p. 1466, 2016.
- [5] D. Magrin, "Network level performances of a LoRa system," 2016.
- [6] L. Alliance, "LoRaWAN-what is it?" *A technical overview of LoRa and LoRaWAN*, 2015.
- [7] F. Adelantado, X. Vilajosana, P. Tuset-Peiro, B. Martinez, J. Melia-Segui, and T. Watteyne, "Understanding the limits of LoRaWAN," *IEEE Communications magazine*, vol. 55, no. 9, pp. 34–40, 2017.
- [8] A. Liljegren and R. Franksson, "Measuring a LoRa network: Performance, possibilities and limitations," 2018.
- [9] K. Mikhaylov, J. Petaejaeravi, and T. Haenninen, "Analysis of capacity and scalability of the LoRa low power wide area network technology," in *Proc. of European Wireless*, 2016, pp. 1–6.
- [10] P. Neumann, J. Montavont, and T. Noël, "Indoor deployment of low-power wide area networks (LPWAN): A LoRaWAN case study," in *Proc. of IEEE WiMob*, 2016, pp. 1–8.
- [11] T.-H. To and A. Duda, "Simulation of lora in ns-3: Improving LoRa performance with CSMA," in *Proc. of IEEE ICC*. IEEE, 2018, pp. 1–7.
- [12] L. M. Feeney and M. Nilsson, "Investigating the energy consumption of a wireless network interface in an ad hoc networking environment," in *Proc. of IEEE INFOCOM*, vol. 3, 2001, pp. 1548–1557.
- [13] L. M. Feeney, "An energy consumption model for performance analysis of routing protocols for mobile ad hoc networks," *Mobile Networks and applications*, vol. 6, no. 3, pp. 239–249, 2001.
- [14] G. Holland and N. Vaidya, "Analysis of TCP performance over mobile ad hoc networks," *Wireless Networks*, vol. 8, no. 2-3, pp. 275–288, 2002.
- [15] J. Haxhibeqiri, F. Van den Abeele, I. Moerman, and J. Hoebeke, "LoRa scalability: A simulation model based on interference measurements," *Sensors*, vol. 17, no. 6, p. 1193, 2017.
- [16] "Rfm95/96/97/98(w) - low power long range transceiver module," https://cdn.sparkfun.com/assets/learn_tutorials/8/0/4/RFM95_96_97_98W.pdf, accessed: July 2019.
- [17] "LoRa energy model," <https://github.com/klimiashvilig/LoRaEnergy>.
- [18] T. Clausen and P. Jacquet, "Optimized link state routing protocol (olsr)," Tech. Rep., 2003.